Worksheet 6

Statistical indices – EXERCISES

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Exercise 1 – Elementary indices, circularity, and reversibility

A service costs 80 euros in 2023, 92 euros in 2024, and 101.20 euros in 2025. Calculate $I_{2024/2023}$, $I_{2025/2024}$, and $I_{2025/2023}$ (base 100); verify circularity; calculate the reverse index $I_{2023/2025}$ and verify reversibility; interpret indices above and below 100. **Exercise 2 – Laspeyres, Paasche, Fisher, and value index**

	2005 (0)		2011 (1)	
Product	p_0	q_0	p_1	q_1
A	53	20	62	25
B	41	28	52	34
C	75	35	82	38
D	25	30	38	33

Calculate the four aggregates, Laspeyres and Paasche price/quantity indices, Fisher price/quantity indices, and the value index. Interpret F^p, F^q, and I_v. **Exercise 3 – Additional application**

For goods (p_0, q_0, p_1, q_1) equal to $(100, 14, 150, 10)$, $(60, 10, 50, 12)$, and $(160, 4, 140, 5)$, calculate the Laspeyres and Paasche price indices and the Fisher price index.